Pollination is the movement of pollen grains from an anther to a stigma. Bees visiting flowers can carry pollen between plants while collecting nectar. Wind also spreads pollen for many grasses and some trees.

After compatible pollen lands on a stigma, it can germinate and grow a pollen tube down the style. The tube carries a male cell toward an ovule. Fertilization occurs when that cell joins the egg; it is a later step than pollination.

Xylem carries water and dissolved minerals from roots toward leaves, mainly through dead vessel elements. Phloem distributes sugars made by photosynthesis to growing or storage tissues. The two transport systems move different materials through a plant.

Stomata are small pores in the leaf surface that allow carbon dioxide to enter and water vapor to escape. Guard cells open or close each pore by changing their pressure. This control balances gas exchange with water loss through transpiration.

C4 plants first fix carbon dioxide into a four-carbon compound in mesophyll cells. They then concentrate carbon dioxide around the enzyme Rubisco in bundle-sheath cells. This pathway can reduce photorespiration in hot, bright conditions.

Seed dispersal moves seeds away from the parent plant. Wind can carry lightweight seeds, while animals may transport seeds attached to fur or after eating fruit. Dispersal reduces crowding and can help plants colonize new habitats.